PUNJAB POLICE • PSSSB • PUNJAB PCS • GOVERNMENT EXAMS

HOUSE OF ASPIRANTS

COMPUTER

CHAPTER 9B

Secondary Memory

Prepared by someone who has already achieved what you are preparing for.

WEBSITE houseofaspirants.in

EDITION October 2026

LANGUAGE Punjabi & English

House of Aspirants

Punjab's bilingual exam-preparation platform

houseofaspirants.in

Welcome, Aspirant

Thank you for choosing **House of Aspirants**. These notes are written for every serious government-exam aspirant: **exam-oriented, easy to understand and free**.

> *"Prepared by someone who has already achieved what you are preparing for."*

HOW TO GET THE MOST FROM THESE NOTES

1 Read
Go through the chapter once and mark the points that are new to you.

2 Practise
Attempt this chapter's MCQ quiz on houseofaspirants.in.

3 Revise
Check previous year questions on the topic and revise your marked points.

Visit Our Website

- Bilingual notes: Punjabi & English
- Daily quiz & chapter-wise MCQs
- Monthly current affairs
- 10+ years of previous year papers
- Progress tracking

Scan to visit

www.houseofaspirants.in

Join Our Telegram Channel

- Latest notes, first
- Daily updates
- Exam notifications
- Weekly PDFs
- Live guidance sessions

Scan to join

t.me/HouseOfAspirant

Keep Learning. Keep Growing. Keep Achieving.

Team House of Aspirants

© 2026 House of Aspirants. These notes are for educational purposes only. Reproduction, redistribution or commercial use without permission is prohibited.

COMPUTER · CHAPTER 9B

Secondary Memory

ਸੈਕੰਡਰੀ ਮੈਮੋਰੀ

IN THIS CHAPTER

- Secondary Memory (Auxiliary Storage)
- Hard Disk Drive (Hdd)
- Ssd (Solid State Drive)
- Magnetic Tape
- Pen Drive
- Memory Card (Sd Card)

SECONDARY MEMORY (AUXILIARY STORAGE)

Secondary Memory refers to storage devices that store data permanently. These devices may be built into the computer or connected externally.

Secondary Memory is also known as:

• Auxiliary Memory

• External Memory

• Backup Storage

Unlike Primary Memory (RAM), Secondary Memory retains data even when the power is switched off.

Features of Secondary Memory

• Non-Volatile Memory

• Large Storage Capacity

• Permanent Data Storage

• Lower Cost Per GB

• Slower than Primary Memory

• CPU Cannot Access Directly

Important Exam Point

The CPU cannot directly access secondary memory. Data must first be transferred to RAM and then processed by the CPU.

CLASSIFICATION OF SECONDARY MEMORY

Secondary Memory | ├── Magnetic Storage or Magnetic Disk | ├── Hard Disk | ├── Floppy Disk | ├── Magnetic Tape | └── ZIP Disk | ├── Solid State Storage | ├── SSD | ├── Pen Drive | └── Memory Card | └── Optical Storage ├── CD ├── DVD ├── Blu-ray Disc └── HVD

HARD DISK DRIVE (HDD)

Hard Disk Drive (HDD) is a magnetic storage device used to store large amounts of data permanently.

It is also called:

- Hard Disk
- Winchester Disk

Common Capacities

- 500 GB
- 1 TB
- 2 TB
- 4 TB and above.

STRUCTURE OF HARD DISK

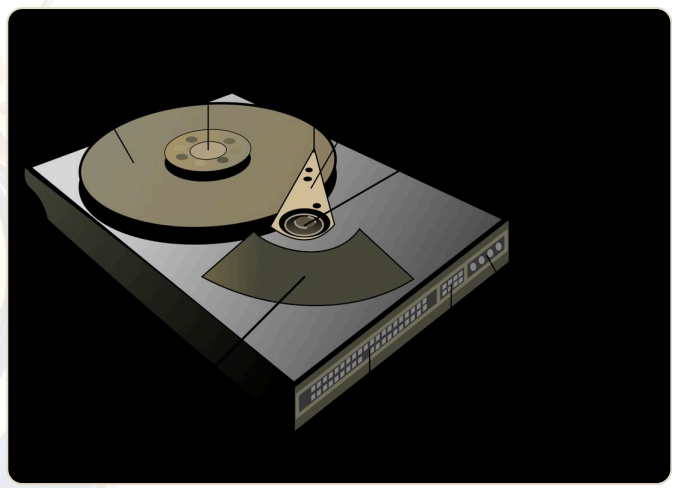
A hard disk consists of several circular magnetic plates called Platters.

Data is stored on both sides of platters.

Speed of hard disk measured in revolution per minute

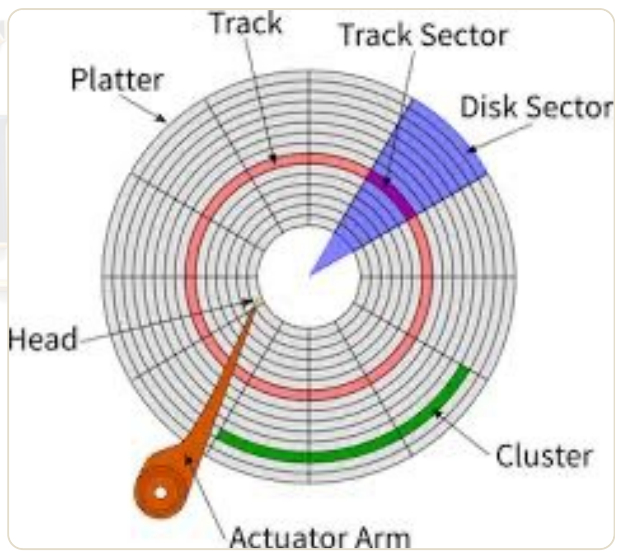
Main Components

- Platter
- Read/Write Head
- Actuator Arm
- Spindle
- Disk Controller Board



DATA ORGANIZATION IN HARD DISK

Platter | ├── Tracks | └── Sectors



Track

Circular paths on the disk surface where data is stored.

Sector

Small subdivision of a track.

Important Exam Point

Traditionally:

1 Sector = 512 Bytes = 2^9 Bytes

(Modern drives may use 4 KB sectors.)

CYLINDER

Tracks having the same radius on all platters together form a Cylinder.

FORMATTING

The process of dividing a disk into tracks and sectors is called Formatting.

Formatting helps the operating system locate and manage files efficiently.

HARD DISK PERFORMANCE TERMS

1. Seek Time

Time taken by the read/write head to reach the required track.

Important Exam Point

Lower Seek Time = Faster Disk

2. Rotational Latency

Time taken by the disk to rotate the required sector under the read/write head.

3. Data Transfer Time

Time taken to transfer data from the disk to memory.

HARD DISK INTERFACES

Hard disks connect to the motherboard through:

1. SATA

Serial Advanced Technology Attachment

Most commonly used interface.

2. PATA

Parallel Advanced Technology Attachment

Older technology.

3. SCSI

Small Computer System Interface

Used in servers and high-performance systems.

HARD DISK ADDRESSING

Addressing refers to locating data on a disk.

1. CHS Addressing

Cylinder – Head – Sector Addressing

Also known as Disk Geometry.

2. LBA Addressing

Logical Block Addressing

Modern method used by operating systems.

Important Exam Point

LBA has largely replaced CHS addressing.

SSD (SOLID STATE DRIVE)

It is a non-volatile storage device that uses Flash Memory instead of magnetic platters.



ADVANTAGES OF SSD

- Faster than HDD
- No Moving Parts
- Silent Operation
- Lower Power Consumption
- More Reliable
- Shock Resistant

Important Exam Point

SSD is significantly faster than HDD because it has no moving mechanical parts.

HDD VS SSD

Feature	HDD	SSD
Technology	Magnetic	Flash Memory
Speed	Slower	Faster
Moving Parts	Yes	No
Noise	Present	Silent
Power Usage	Higher	Lower
Cost	Cheaper	Costlier

MAGNETIC TAPE



Magnetic Tape is a secondary storage device primarily used for:

- Data Backup
- Archival Storage
- Large Data Sets

The concept was developed by Fritz Pfleumer in 1928 in Germany.

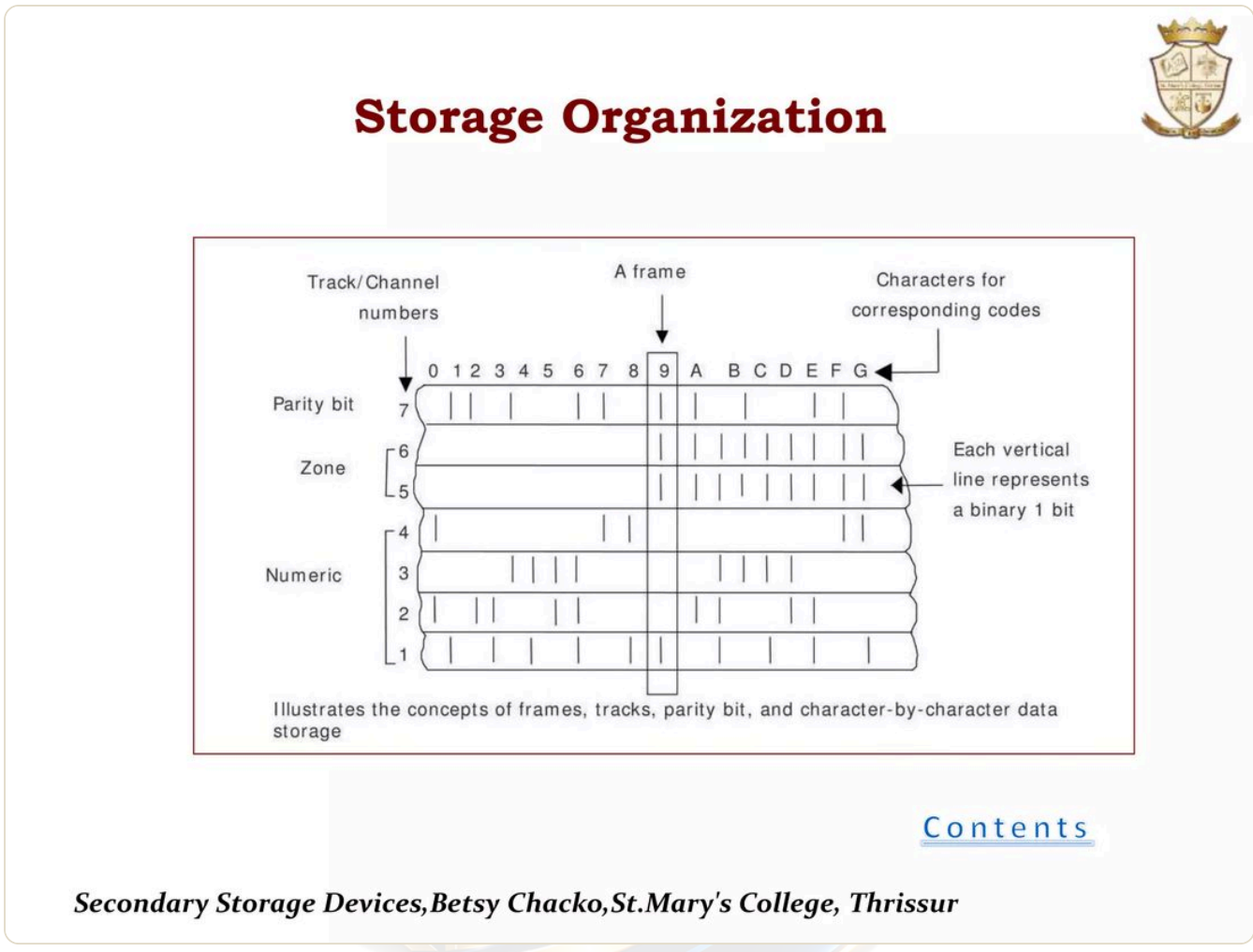
Tape is made up of a plastic ribbon, usually half an inch or a quarter inch wide and 50 to 2400 feet long.

The surface of the tape is coated with iron oxide or chromium dioxide.

It records data by magnetization, can store and erase data, and can read the data several times.

FEATURES OF MAGNETIC TAPE

- Sequential Access
- Low Cost
- Large Capacity
- Used for Backup
- Non-Volatile



STRUCTURE OF MAGNETIC TAPE

Data in magnetic tape is recorded serially and has no addressing of storage.

There are parallel tracks on the tape called tracks, and the number of parallel tracks present on the tape is where data is stored.

Modern magnetic tape has nine tracks and uses an 8-bit EBCDIC code format.

In magnetic tape, data is called records.

The character bit is stored as a pattern of magnetic spots across the width of the tape, and the magnetic spots are recorded on 8 tracks.

An extra bit called a parity bit is stored on the 9th track, called the parity track.

Data on the tape is stored in groups called data blocks, each data block has a gap between them called an inter-block gap (IBG).

A vertical column in the tape is called a frame.

Each record (data) on the tape is separated by a gap that is called an inter-record gap (IRG) and each IRG has a length of 0.5 inches.

The standard size of tape is 1 inch, 1/2 inch, or 1/4 inch.

Data is recorded on the tape in binary 0 or 1.

The device used to read or write on the tape is called a tape streamer.

Blocking is the process of grouping two or more records to form a block of data, and the number of records grouped in each block is called the blocking factor.

Data recorded density is the number of bytes of records per inch.

In all magnetic tape, data is read or written serially, not randomly.

Magnetic tape drives, which are used to read or write data in magnetic tape, have two tape reels: one to read data called the file reel and the other to write data called the take-up reel.

Inter-record gaps separate records by blank spaces, and inter-block gaps separate blocks by blank spaces.



TYPES OF MAGNETIC TAPE

- 1/2 Inch Tape Reel
- Tape Cartridge
- Streamer Tape
- Digital Audio Tape (DAT)

IMPORTANT EXAM FACT

Magnetic Tape uses Sequential Access only.

PEN DRIVE

Pen Drive is a portable flash memory device.

Other Names:

- USB Flash Drive
- Thumb Drive
- Jump Drive

FEATURES

- Plug and Play
- Portable
- Reusable
- Non-Volatile
- USB Interface

Common Capacity

2 GB to 1 TB and above.

MEMORY CARD (SD CARD)

SD Stands for Secure Digital Card

Used in:



- Smartphones
- Cameras
- Tablets

Types

- Standard SD
  - Mini SD
  - Micro SD

CD (COMPACT DISC)

Developed by Philips and Sony in 1982.

A standard CD is around 5 inches in diameter and 0.05 inches in thickness



Used for storing:

- Audio
- Video
- Software
- Documents

Capacity :Approximately 700 MB

TYPES OF CD

CD-worm(write once read many)

CD-ROM :Read Only Memory

CD-R :Record Once

CD-RW :Rewritable Disc

DVD (DIGITAL VERSATILE DISC)

DVD stores much more data than a CD.

DVD was co-developed by Sony, Panasonic, Philips, and Toshiba in 1995.

TYPES OF DVD

- DVD-ROM :Read Only
- DVD-R :Write Once
- DVD-RW :Rewritable
- BLU-RAY DISC (BD)
  - Uses Blue Laser Technology.
  - Capacity: 25 GB (Single Layer) ,50 GB (Dual Layer)



Important Correction

Blu-ray capacity is much larger than 2.5 GB.

- HVD :Holographic Versatile Disc
  - Capacity :Approximately 3.9 TB
  - Used for very high-capacity storage.

FLASH MEMORY

Flash Memory is a non-volatile semiconductor memory.

Data remains stored even after power is switched off.

USES OF FLASH MEMORY

• SSD

• Pen Drive

• Memory Cards

• Smartphones

• Digital Cameras

• Set Top Boxes

ADVANTAGES

• Fast Access

• Reliable

• Portable

• Low Power Consumption

VIRTUAL MEMORY

Virtual Memory is a technique that uses part of the hard disk/SSD as temporary RAM.

When RAM becomes full, the operating system moves less-used data from RAM to storage.

Important Exam Point

Virtual Memory increases apparent memory but is slower than RAM.

MEMORY ACCESS METHODS

Memory Access refers to the process by which a computer system reads data from or writes data to its memory. Memory access is a fundamental operation in computing, as it allows the processor to retrieve instructions and data needed to execute programs. There are two main types of memory access:

1. Read: Fetching data or instructions from memory.

2. Write : Storing data or results into memory.

These are 4 types of memory access methods:

1. SEQUENTIAL ACCESS

Data is accessed in linear sequential manner order.

Example:Magnetic Tape

2. RANDOM ACCESS

Any memory location can be accessed directly.

Examples:RAM, ROM

3. DIRECT ACCESS

Combination of sequential and random access.

Examples:

- Hard Disk
- Optical Disk
- Flash Drive

4. ASSOCIATIVE ACCESS

Data is accessed by content rather than address.

Example:

Cache Memory

MEMORY UNITS

Memory units are used to measure and represent data. These units, which can be measured in bits and bytes, offer a technique to estimate the volume of data which can be stored in a specific device or system. Some of the commonly used memory units are:

Bit

Smallest unit of data.

1 Bit = 0 or 1

Nibble

1 Nibble = 4 Bits

Byte

1 Byte = 8 Bits

Kilobyte (KB)

Full Form	Units	Bytes
1 Bit	Binary Digit (0/1)	
1 Nibble	4 bits	
1 Byte	8 bits	
1 kilobyte(KB)	1024 byte	2^{10} bytes
1 Megabyte(MB)	1024 KB	2^{20} bytes
1 Gigabyte (GB)	1024 MB	2^{30} bytes
1 Terabyte(TB)	1024 GB	2^{40} bytes
1 Petabyte(PB)	1024 TB	2^{50} bytes
1 Exabyte(EB)	1024 PB	2^{60} bytes
1 Zettabyte(ZB)	1024 EB	2^{70} bytes
1 Yottabyte(YB)	1024 ZB	2^{80} bytes
1 Brontobyte	1024 YB	2^{90} bytes
1 Geopbyte	1024 Brontobyte	2^{100} bytes

1 KB = 1024 Bytes

Megabyte (MB)

1 MB = 1024 KB

Gigabyte (GB)

1 GB = 1024 MB

Terabyte (TB)

1 TB = 1024 GB

Petabyte (PB)

1 PB = 1024 TB

Exabyte (EB)

1 EB = 1024 PB

Zettabyte (ZB)

1 ZB = 1024 EB

MEMORY SIZES

Register :bytes

Cache memory :kb/mb

RAM:GB

USB:GB

Magnetic disk/hard disk:TB

Magnetic tapes:PB

QUICK REVISION NOTES

- ✔ Secondary Memory is non-volatile.
- ✔ HDD is magnetic storage.
- ✔ SSD uses flash memory.
- ✔ HDD contains tracks and sectors.
- ✔ Cylinder = Tracks of same radius.
- ✔ Formatting divides disk into tracks and sectors.
- ✔ Magnetic Tape uses sequential access.
- ✔ Pen Drive uses flash memory.
- ✔ Blu-ray uses blue laser.
- ✔ Virtual Memory uses hard disk space.
- ✔ Cache uses associative access.
- ✔ RAM uses random access.

MOST IMPORTANT FACTS FOR COMPETITIVE EXAMS

1. HDD is also called Winchester Disk.

2. SSD has no moving parts.

3. 1 Sector = 512 Bytes (traditional).

4. CHS = Cylinder Head Sector.

5. LBA = Logical Block Addressing.

6. Magnetic Tape uses Sequential Access.

7. Cache Memory uses Associative Access.

8. RAM uses Random Access.

9. DVD stores more data than CD.

10. Flash Memory is used in SSDs and Pen Drives.

ONE-LINE REVISION POINTS

• HDD stores data magnetically.

- SSD uses flash memory.
- Cylinder = Tracks of same radius.
- Seek Time affects HDD speed.
- SATA is the most common HDD interface.
- Magnetic Tape is best for backup.
- Pen Drive is a portable flash storage device.
- SD stands for Secure Digital.
- Blu-ray uses blue laser technology.
- Virtual Memory uses disk space as RAM.
- Sequential Access = Tape.
- Random Access = RAM.
- Associative Access = Cache.
- 1 Byte = 8 Bits.
- 1 TB = 1024 GB
- Memory Types Speed Order:

Register → Cache → RAM → SSD → HDD → Tape

(Fastest → Slowest)